



AUTOMATIC NUMBER PLATE RECOGNITION | DAHEJ LNG TERMINAL

Every Vehicle Seen. Every Movement Recorded.

A professional case study and improvement concept covering capture geometry, ANPR workflow, car/truck detection points, reporting, reliability and site beautification.

OPERATIONAL CHANNELS

4 ANPR Cameras

RECOGNITION TARGET

Above 98%

EVIDENCE SAMPLE

33 Records / Hour

Capture. Read. Classify. Verify. Report.

Main Gate and Parking Gate coverage with central servers, VMS integration and operator reporting.
Order 4400000622 | Operational evidence dated 16 May 2024 | Commercial values excluded.

Project Evidence and Outcome



PROJECT MANDATE

Automate vehicle identity at critical entry and exit control points.

D-TECH supplied and integrated an ANPR platform for Petronet LNG at Dahej, covering cameras, IR, servers, storage, VMS, network, civil works, commissioning, training and support.

ORDER DATE

19 Feb 2024

COMPLETION TARGET

15 May 2024

OPERATIONAL EVIDENCE

16 May 2024

Verified project proof

● Installed field infrastructure

Dual-arm pole arrangements and ANPR camera positions are visible in supplied photographs.

● Multiple control points

Records identify Main Gate and Parking Gate entry/exit camera channels.

● System-generated records

The supplied one-hour LPR report contains 33 detection records.

● Vehicle and plate evidence

Each report event carries time, camera, vehicle image and extracted plate crop.

What D-TECH Delivered

A complete ANPR system - from roadside capture to report export.

The composite order defines a turnkey supply, installation and integration scope. The quantities and capabilities below are drawn from the technical specification and price schedule.

01

ANPR CAPTURE

4 operational bullet cameras plus 1 spare; deep-learning analytics and IR capability.

02

NIGHT ILLUMINATION

4 dedicated IR illuminators with adjustable coverage and outdoor protection.

03

COMPUTE + STORAGE

Application server and 4-channel storage server designed for 60-day retention.

04

VMS INTEGRATION

4 VMS channel licences, analytics integration and connection to existing CCTV.

05

OPERATOR DESK

Workstation, accessories and 32-inch live-view display.

06

FIELD INFRASTRUCTURE

4 three-metre poles, IP65 junction boxes, racks, switches, fibre and power cabling.

07

SOFTWARE PLATFORM

ANPR dashboard, search, alerts, correction, multi-entry console and report exports.

08

DELIVERY SERVICES

Survey, design, civil/electrical works, testing, tuning, training and documentation.

What Must Be Controlled for Reliable ANPR

Recognition accuracy depends on the complete capture chain.

The software target was above 98%, but real performance depends on camera angle, plate size, illumination, lane discipline, time synchronization, network health and reporting rules. These are the key controls for dependable results.

01

Camera geometry

Avoid excessive horizontal or vertical angle; dedicate the view to a defined plate-capture zone.

02

Vehicle path

Channel cars, buses and trucks through predictable lane centres and controlled approach speed.

03

Lighting balance

Use IR to control night capture while avoiding reflective glare and overexposure.

04

Time alignment

Synchronize camera, server, VMS and report timestamps from one approved time source.

05

Direction logic

Configure entry/exit direction, duplicate suppression and correct gate association.

06

Exception handling

Provide plate correction, unlisted alerts, unreadable-event review and audit trail.

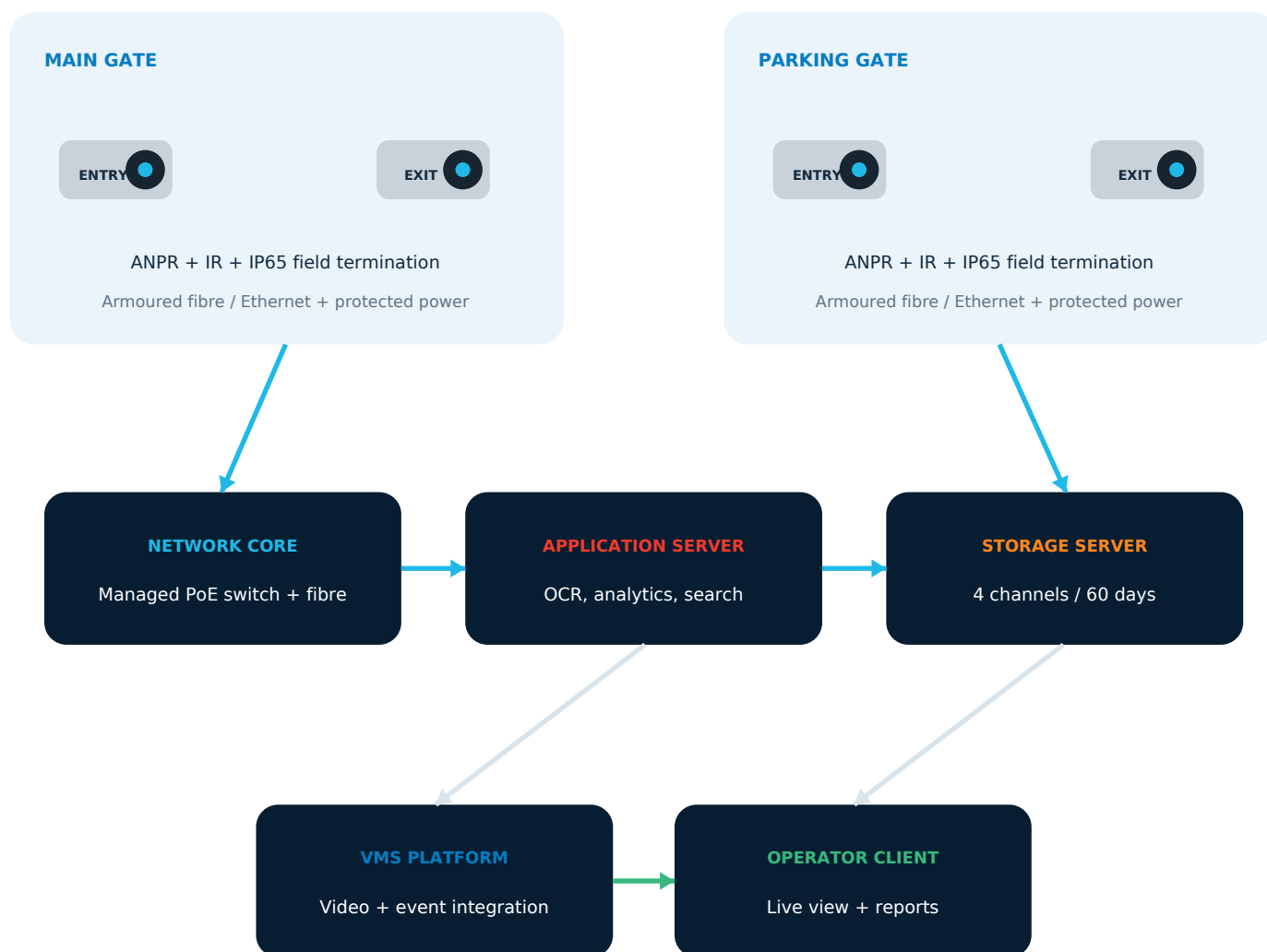
REPORT RECOMMENDATION

Validate each channel with a timed physical vehicle log across day/night and compare capture time, plate result, direction and gate against the ANPR event record. Correct geometry or rules before accepting the KPI baseline.

Multi-Gate ANPR Architecture

Four capture channels feed one searchable security platform.

The conceptual architecture below reflects the ordered components. Exact port allocation, IP addressing and VLAN design should follow the approved as-built network diagram.



DATA FLOW

Vehicle image + plate crop + OCR text + class + gate + direction + timestamp + match-list status

Events remain searchable through the application and linked to recorded video for verification and export.

Car and Truck Sample Detection Points

One lane design must handle very different vehicle geometry.

Cars and trucks present plates at different heights, approach angles and positions. The diagrams below are conceptual commissioning guides; final distances and angles must be tuned on site to the selected camera, lens and software.

SAMPLE POINT A - CAR / LIGHT VEHICLE



Plate capture line

Aim at the lane centre and plate plane.
Minimize skew, glare and background clutter.



SAMPLE POINT B - TRUCK / LNG TANKER



Dedicated heavy-vehicle focus

Account for higher plate location and long body.
Avoid cab shadow and adjacent-lane interference.



Recommended commissioning principle: one clearly defined capture zone per lane, verified with both front and rear plate styles.

From Vehicle Frame to Searchable Record

The event is useful only when image, text and time agree.

The ANPR pipeline combines visual evidence with machine-readable data. Every stage should retain enough context for the operator to verify or correct the result.

01 CAPTURE

Vehicle enters the configured detection zone.

02 DETECT

Analytics identifies vehicle class and movement.

03 CROP

System isolates and enhances the plate region.

04 OCR

Characters are translated into registration text.

05 MATCH

Result is checked against list / database rules.

06 LOG

Images, text, gate, direction and time are stored.

ANONYMIZED SAMPLE EVENT



TIMESTAMP	16 May 2024 17:15:31
CAMERA	Parking Gate Entry/Exit 1
VEHICLE TYPE	Car / Light Vehicle
REGISTRATION	GJ 16 XX 1234 (anonymized)
MATCH STATUS	Unlisted - operator review

Sample mirrors the supplied LPR report fields. Registration has been masked for client-safe presentation.

Reports That Support Security Decisions

Move from event lists to measurable gate performance.

The ordered system supports custom reports and exports in PDF, Excel and Word. A useful operating dashboard should combine volume, quality, exceptions and system health.

TOTAL DETECTIONS

33

1-hour supplied sample

CAMERA GROUPS

2

Main + Parking Gate

TARGET ACCURACY

>98%

Contract requirement

STORAGE BASIS

60 Days

Four channels

RECOMMENDED KPI SET

Read rate

Readable plates / total vehicle passages, by channel and shift.

Correct-read rate

Verified correct OCR / readable plate events.

Missed passage rate

Physical passages with no corresponding ANPR event.

Duplicate rate

Repeated events for one passage inside the suppression window.

Time variance

Difference between physical/VMS time and ANPR report time.

Lane attribution

Events assigned to the correct gate, lane and direction.

System availability

Camera, server, storage and application uptime.

Exception ageing

Time taken to review unreadable, unlisted or corrected events.

DATA QUALITY RULE

A recognition KPI is valid only when compared against an independent, time-aligned physical or video ground truth.

ANPR Pole and Gate Beautification

Improve appearance without compromising capture geometry.

The current installation proves functionality. A structured physical upgrade can improve presentation, maintainability and protection while preserving the validated camera aim.



RECOMMENDED PHYSICAL IMPROVEMENTS

- Galvanized finished structure**
Replace exposed fabricated members with a powder-coated, engineered dual-arm pole or compact gantry.
- Serviceable IP enclosure**
Place lockable IP-rated junction and isolation points at safe maintenance height.
- Lane identity and signage**
Mark Entry/Exit, camera ID and ANPR zone; add speed and lane-centre guidance.
- Concealed cable routing**
Use closed conduits/trays, sealed glands and labelled service loops; remove visible loose cabling.
- Protected pole base**
Provide finished RCC plinth, protective bollards, reflective chevrons and drainage clearance.
- Safety and access**
Add anti-climb control, lockable ladder access and approved maintenance platform/anchor points.

DESIGN CONTROL

Do not move a validated camera solely for appearance. Re-survey and re-test every channel after structural changes.

Acceptance, Support and Business Value



PETRONET LNG | DAHEJ LNG TERMINAL

A traceable vehicle-security layer built for continuous operations.

The project combines field capture, analytics, server/storage infrastructure, VMS integration and reporting in one maintainable solution.

01

BASELINE

Physical count + ANPR export + video review for every channel.

02

TUNE

Angle, zoom, IR, OCR region, direction and duplicate rules.

03

ACCEPT

Day/night car and truck trials with signed KPI evidence.

04

OPERATE

Daily health check, exception review and monthly KPI report.

05

IMPROVE

Beautification, lifecycle review and controlled change testing.

Business value

● Stronger gate control

Every detected passage becomes searchable evidence.

● Faster investigation

Operators search by plate, time, gate and event.

● Lower manual logging

Automated capture reduces repetitive gate-register work.

● Scalable architecture

Multiple entry points operate from a common console.

D-TECH | INTELLIGENT GATES. VERIFIED MOVEMENT. ACTIONABLE SECURITY.